

Chapter 2 – Graph Theory

Formula Sheet

Network Science · Barabasi et al.

Master Symbol Reference

N	Number of nodes (network size)	L	Number of links
i, j	Node indices ($i = 1, 2, \dots, N$)	k_i	Degree of node i
$\langle k \rangle$	Average (mean) degree	p_k	Degree distribution
N_k	Number of nodes with degree k	A_{ij}	Adjacency matrix element
w_{ij}	Weight of link (i, j)	B_{ie}	Incidence matrix element
k_i^{in}	In-degree of node i	k_i^{out}	Out-degree of node i
L_i	Number of links among neighbours of i	C_i	Local clustering coefficient
$\langle C \rangle$	Average clustering coefficient	C_Δ	Global clustering coefficient
d_{ij}	Shortest path length (distance) between i and j	d_{max}	Network diameter
$\langle d \rangle$	Average path length	L_{max}	Max possible links
$\langle x \rangle$	Mean (average) of quantity x	σ_x	Standard deviation of x

1. Links and Degrees

Total Number of Links (Undirected Network)

$$L = \frac{1}{2} \sum_{i=1}^N k_i \quad (2.1)$$

L	Total number of links in the network
k_i	Degree of node i – the number of links attached to node i
N	Total number of nodes
$\frac{1}{2}$	Correction factor: each link connects two nodes, so it is counted once in each node's degree, hence summing all degrees double-counts every link

Average Degree (Undirected Network)

$$\langle k \rangle = \frac{1}{N} \sum_{i=1}^N k_i = \frac{2L}{N} \quad (2.2)$$

$\langle k \rangle$	Average (mean) degree of all nodes in the network
k_i	Degree of node i
N	Total number of nodes
L	Total number of links; the factor of 2 arises from equation (2.1)

Total Degree of a Node (Directed Network)

$$k_i = k_i^{\text{in}} + k_i^{\text{out}} \quad (2.3)$$

k_i	Total degree of node i : the combined count of all its connections
k_i^{in}	In-degree: number of directed links pointing <i>towards</i> node i
k_i^{out}	Out-degree: number of directed links pointing <i>away from</i> node i

Total Number of Links (Directed Network)

$$L = \sum_{i=1}^N k_i^{\text{in}} = \sum_{i=1}^N k_i^{\text{out}} \quad (2.4)$$

L	Total number of directed links in the network
k_i^{in}	In-degree of node i : each directed link contributes 1 to exactly one node's in-degree ...
k_i^{out}	... and 1 to exactly one node's out-degree, so both sums equal L (no double-counting, hence no $\frac{1}{2}$ factor)

Average Degree (Directed Network)

$$\langle k^{\text{in}} \rangle = \frac{1}{N} \sum_{i=1}^N k_i^{\text{in}} = \langle k^{\text{out}} \rangle = \frac{1}{N} \sum_{i=1}^N k_i^{\text{out}} = \frac{L}{N} \quad (2.5)$$

$\langle k^{\text{in}} \rangle$	Average in-degree across all nodes
$\langle k^{\text{out}} \rangle$	Average out-degree across all nodes
L	Total number of directed links
N	Total number of nodes
The average in-degree always equals the average out-degree, because every link contributes exactly one unit to each.	

2. Degree Distribution

Degree Distribution

$$p_k = \frac{N_k}{N} \quad (2.7)$$

p_k	Probability that a randomly chosen node has exactly k links
N_k	Number of nodes in the network whose degree is exactly k
N	Total number of nodes

Normalisation Condition

$$\sum_{k=0}^{\infty} p_k = 1 \quad (2.6)$$

p_k	Degree distribution: the probability of a node having degree k This states that every node must have <i>some</i> degree; the probabilities over all possible degrees must sum to 1.
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Average Degree via Degree Distribution

$$\langle k \rangle = \sum_{k=0}^{\infty} k p_k \quad (2.8)$$

$\langle k \rangle$	Average degree of the network
k	A specific degree value (number of links)
p_k	Probability that a randomly chosen node has degree k This is the standard formula for the mean of a discrete probability distribution: multiply each degree value by its probability and sum over all possible degrees.

3. Adjacency Matrix

Degree from Adjacency Matrix (Undirected)

$$k_i = \sum_{j=1}^N A_{ji} = \sum_{j=1}^N A_{ij} \quad (2.9)$$

k_i	Degree of node i
A_{ij}	Adjacency matrix element: equals 1 if there is a link from node j to node i , else 0

Because the adjacency matrix is symmetric for undirected networks ($A_{ij} = A_{ji}$), summing either the i -th row or the i -th column gives the degree of node i .

In-degree and Out-degree from Adjacency Matrix (Directed)

$$k_i^{\text{in}} = \sum_{j=1}^N A_{ij}, \quad k_i^{\text{out}} = \sum_{j=1}^N A_{ji} \quad (2.10)$$

k_i^{in}	In-degree of node i : obtained by summing the i -th <i>row</i> of A
k_i^{out}	Out-degree of node i : obtained by summing the i -th <i>column</i> of A
A_{ij}	Adjacency matrix element: 1 if a link points from j to i , else 0

Total Number of Non-zero Elements (Undirected)

$$2L = \sum_{i=1}^N k_i^{\text{in}} = \sum_{i=1}^N k_i^{\text{out}} = \sum_{i,j} A_{ij} \quad (2.11)$$

L	Total number of links
A_{ij}	Adjacency matrix element (0 or 1 for unweighted networks)
$2L$	The count of non-zero entries in the adjacency matrix, since each undirected link (i, j) appears as both $A_{ij} = 1$ and $A_{ji} = 1$

Weighted Adjacency Matrix

$$A_{ij} = w_{ij} \quad (2.13)$$

A_{ij}	Element of the adjacency matrix for node pair (i, j)
w_{ij}	Weight of the link between nodes i and j (e.g. minutes of phone calls, current flow)
For unweighted (binary) networks $w_{ij} \in \{0, 1\}$; for weighted networks w_{ij} can be any non-negative real number.	

4. Network Size and Sparsity

Maximum Number of Links (Complete Graph)

$$L_{\text{max}} = \binom{N}{2} = \frac{N(N-1)}{2} \quad (2.12)$$

L_{max}	Maximum possible number of links in a network of N nodes (achieved in a complete graph)
N	Total number of nodes
$\binom{N}{2}$	Binomial coefficient: the number of ways to choose 2 nodes from N , each choice corresponding to one possible link

Sparsity check: A network is sparse if $L \ll L_{\max}$, i.e. $L \ll N(N-1)/2$. For example, the WWW has $\approx 1.5 \times 10^6$ links but $L_{\max} \approx 5 \times 10^{10}$: only 3×10^{-5} of possible links are present.

5. Paths and Number of Shortest Paths

Number of Paths of Length d Between Nodes i and j

$$N_{ij}^{(d)} = [A^d]_{ij} \text{ (Box 2.4)}$$

$N_{ij}^{(d)}$	Number of distinct paths of exactly d steps between nodes i and j
A^d	The adjacency matrix A raised to the power d (i.e. matrix multiplied by itself d times)
$[A^d]_{ij}$	The (i, j) element of the matrix A^d
d	Path length: the number of links in the path

Finding the distance: The shortest path distance d_{ij} is the smallest integer d for which $N_{ij}^{(d)} > 0$, i.e. $[A^d]_{ij} > 0$.

Average Path Length (Directed Network)

$$\langle d \rangle = \frac{1}{N(N-1)} \sum_{\substack{i,j=1 \\ i \neq j}}^N d_{ij} \text{ (2.14)}$$

$\langle d \rangle$	Average path length: the mean distance between all pairs of nodes
d_{ij}	Shortest path distance between nodes i and j
$N(N-1)$	Total number of ordered node pairs (i, j) with $i \neq j$ in a directed network of N nodes

For an **undirected** network: $\langle d \rangle = \frac{1}{\frac{N(N-1)}{2}} \sum_{i < j} d_{ij}$, since $d_{ij} = d_{ji}$. This formula is computed only for node pairs within the same connected component.

6. Clustering Coefficient

Local Clustering Coefficient

$$C_i = \frac{2 L_i}{k_i (k_i - 1)} \text{ (2.15)}$$

C_i	Local clustering coefficient of node i : a value between 0 and 1
L_i	Number of links that actually exist between the k_i neighbours of node i
k_i	Degree of node i (number of its direct neighbours)
$k_i(k_i - 1)/2$	Maximum possible links among k_i neighbours (denominator before multiplying by 2)
$2 L_i$	Numerator accounts for the fact that each link between neighbours is counted once in L_i but the denominator counts ordered pairs

Interpretation: $C_i = 0$ means none of i 's neighbours are connected to each other. $C_i = 1$ means all neighbours form a complete graph (clique). Probabilistically, C_i is the chance that two randomly selected neighbours of node i are also connected. Undefined (set to 0) when $k_i \leq 1$.

Average Clustering Coefficient

$$\langle C \rangle = \frac{1}{N} \sum_{i=1}^N C_i \quad (2.16)$$

$\langle C \rangle$	Average clustering coefficient of the entire network
C_i	Local clustering coefficient of node i (from Eq. 2.15)
N	Total number of nodes

Interpretation: $\langle C \rangle$ is the probability that two neighbours of a randomly selected node are also connected to each other.

Global Clustering Coefficient

$$C_{\Delta} = \frac{3 \times (\text{number of triangles})}{(\text{number of connected triplets})} \quad (2.17)$$

C_{Δ}	Global clustering coefficient (ratio of transitive triplets)
Triangle	A set of 3 nodes each connected to the other two (a closed triplet)
Connected triplet	An ordered set of 3 nodes A, B, C where $A-B$ and $B-C$ are linked (but $A-C$ need not be)
Factor of 3	Each triangle contributes to 3 different connected triplets (one for each node as the “middle” node), so the numerator corrects for this

$\langle C \rangle$ and C_{Δ} are **not equivalent**. $\langle C \rangle$ weights each node equally; C_{Δ} weights each triplet equally and is dominated by low-degree nodes (which are more numerous).

7. Statistical Quantities (Box 2.2)

Mean (Average)

$$\langle x \rangle = \frac{x_1 + x_2 + \cdots + x_N}{N} = \frac{1}{N} \sum_{i=1}^N x_i$$

$\langle x \rangle$	The arithmetic mean of quantity x over all N observations
x_i	The value of x for the i -th data point or node
N	Total number of observations (nodes)

n -th Moment

$$\langle x^n \rangle = \frac{1}{N} \sum_{i=1}^N x_i^n$$

$\langle x^n \rangle$	The n -th moment of quantity x
n	The order of the moment ($n = 1$ gives the mean; $n = 2$ appears in variance)
x_i	Value of x for the i -th data point

Standard Deviation

$$\sigma_x = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \langle x \rangle)^2} = \sqrt{\langle x^2 \rangle - \langle x \rangle^2}$$

σ_x	Standard deviation: measures spread of x values around their mean
x_i	Value of x for the i -th data point
$\langle x \rangle$	Mean of x
$(x_i - \langle x \rangle)^2$	– Squared deviation of each data point from the mean

Discrete Probability Distribution

$$p_x = \frac{1}{N} \sum_i \delta_{x, x_i}, \quad \text{subject to} \quad \sum_x p_x = 1$$

p_x	Probability that quantity x takes value x
δ_{x, x_i}	Kronecker delta: equals 1 if $x = x_i$, else 0
$\sum_x p_x = 1$	Normalisation: probabilities over all outcomes must sum to 1

8. Metcalfe's Law

Network Value vs. Cost

$$\text{Value} \sim N^2, \quad \text{Cost} \sim N, \quad (\text{Value is profitable when } N \text{ is large})$$

N Number of nodes (users or devices) in the network

N^2 Approximate number of possible pairwise connections $\approx N(N - 1)/2$, driving the value of the network

Since value grows as N^2 while cost grows as N , once N exceeds a critical mass the technology becomes profitable ($N^2 > N$ for $N > 1$).

Limitation: In practice, real networks are *sparse* ($L \ll N^2/2$), so the actual value grows closer to N , not N^2 . Furthermore, links have unequal weights, so not all connections are equally valuable.

9. L and $\langle k \rangle$ for Different Graph Types

The formulas for the number of links L and average degree $\langle k \rangle$ change depending on the type of graph. The six cases below correspond directly to the standard graph types from the textbook.

(a) Undirected Network

$$L = \frac{1}{2} \sum_{i,j=1}^N A_{ij}, \quad \langle k \rangle = \frac{2L}{N}$$

$$A_{ii} = 0 \quad (\text{no self-loops}), \quad A_{ij} = A_{ji} \quad (\text{symmetric})$$

L Number of links; the $\frac{1}{2}$ corrects for each link being counted twice (once per endpoint)

A_{ij} Adjacency matrix element: 1 if nodes i and j are connected, 0 otherwise

$A_{ii} = 0$ Diagonal entries are zero – nodes do not connect to themselves

$A_{ij} = A_{ji}$ Matrix is symmetric – connections are mutual

$\langle k \rangle$ Average degree; factor of 2 because each link contributes to two nodes' degrees

(b) Undirected Network with Self-loops

$$L + S = \frac{1}{2} \sum_{\substack{i,j=1 \\ i \neq j}}^N A_{ij} + \sum_{i=1}^N A_{ii}, \quad \langle k \rangle = \frac{2L + 2S}{N}$$

$$\exists i : A_{ii} \neq 0 \quad (\text{self-loops present}), \quad A_{ij} = A_{ji}$$

$L + S$	Total link count; split into two parts: off-diagonal links (L) and self-loops (S)
$\frac{1}{2} \sum_{i \neq j} A_{ij}$	Counts ordinary links (each shared between two nodes, so halved)
$\sum_{i=1}^N A_{ii}$	Counts self-loops directly – a self-loop belongs entirely to one node, so <i>no</i> halving is applied
$A_{ii} \neq 0$	At least one node connects to itself

The average degree $\langle k \rangle$ for networks with self-loops depends on how degree is defined for self-loops (whether a self-loop counts as 1 or 2 towards a node's degree). This is why the textbook marks it with “?”.

(c) Undirected Multigraph

$$L = \frac{1}{2} \sum_{i,j=1}^N A_{ij}, \quad \langle k \rangle = \frac{2L}{N}$$

$$A_{ii} = 0, \quad A_{ij} = A_{ji}, \quad A_{ij} \in \{0, 1, 2, 3, \dots\}$$

L	Total number of links, counting each parallel link separately
A_{ij}	Now an integer ≥ 0 representing the <i>number</i> of parallel links between nodes i and j (not just 0 or 1 as in a simple graph)
$A_{ii} = 0$	No self-loops in this variant
$A_{ij} = A_{ji}$	Still symmetric – each pair of parallel links is mutual
$\langle k \rangle$	Average degree; same formula as undirected, but k_i now counts all parallel links

(d) Directed Network

$$L = \sum_{i,j=1}^N A_{ij}, \quad \langle k \rangle = \frac{L}{N}$$

$$A_{ij} \neq A_{ji} \quad (\text{asymmetric in general})$$

L	Total number of directed links; no $\frac{1}{2}$ factor because each directed link belongs to one node as out-link and another as in-link – there is no double counting
A_{ij}	Adjacency element: 1 if a directed link points <i>from</i> j <i>to</i> i , else 0
$A_{ij} \neq A_{ji}$	Matrix is asymmetric – a link from j to i does not imply a link from i to j
$\langle k \rangle = L/N$	Average degree is L/N (not $2L/N$) because each link is not shared between two undirected endpoints

(e) Weighted Undirected Network

$$L = \frac{1}{2} \sum_{i,j=1}^N \mathbf{1}[A_{ij} > 0], \quad \langle k \rangle = \frac{2L}{N}$$

$$A_{ii} = 0, \quad A_{ij} = A_{ji} = w_{ij} \in \mathbb{R}^+$$

L	Number of links (structural); counts the number of node pairs with a non-zero weight, regardless of the weight's magnitude
$A_{ij} = w_{ij}$	Matrix entries are real-valued weights, not just 0 or 1
$\mathbf{1}[A_{ij} > 0]$	Indicator function: equals 1 if a link exists (any weight), 0 otherwise
$\langle k \rangle$	Average <i>unweighted</i> degree (link count per node); the weighted analogue is the <i>node strength</i> $s_i = \sum_j w_{ij}$

Node strength: For weighted networks a more informative quantity than degree is the **strength** $s_i = \sum_j A_{ij} = \sum_j w_{ij}$, which sums the weights of all links attached to node i .

(f) Complete Graph (Undirected)

$$L = L_{\max} = \frac{N(N-1)}{2}, \quad \langle k \rangle = N-1$$

$$A_{ii} = 0, \quad A_{i \neq j} = 1 \quad (\text{all off-diagonal entries are 1})$$

$L = L_{\max}$	Every possible link exists; the network has the maximum number of links
$\frac{N(N-1)}{2}$	The number of ways to choose 2 nodes from N (binomial coefficient $\binom{N}{2}$)
$\langle k \rangle = N-1$	Each node is connected to every other node, so degree equals $N-1$
$A_{i \neq j} = 1$	Every off-diagonal entry of the adjacency matrix is 1
$A_{ii} = 0$	Nodes do not connect to themselves

Graph Type	L	$\langle k \rangle$	A_{ii}	$A_{ij} = A_{ji}?$
Undirected	$\frac{1}{2} \sum_{ij} A_{ij}$	$2L/N$	$= 0$	Yes
Self-loops	$\frac{1}{2} \sum_{i \neq j} A_{ij} + \sum_i A_{ii}$	?	$\neq 0$	Yes
Multigraph	$\frac{1}{2} \sum_{ij} A_{ij}$	$2L/N$	$= 0$	Yes
Directed	$\sum_{ij} A_{ij}$	L/N	$= 0$	No
Weighted	$\frac{1}{2} \sum_{ij} \mathbf{1}[A_{ij} > 0]$	$2L/N$	$= 0$	Yes
Complete	$N(N-1)/2$	$N-1$	$= 0$	Yes

10. Quick Reference Summary

Quantity	Formula	Eq.
Links (undirected)	$L = \frac{1}{2} \sum_i k_i$	(2.1)
Avg degree (undirected)	$\langle k \rangle = 2L/N$	(2.2)
Total degree (directed)	$k_i = k_i^{\text{in}} + k_i^{\text{out}}$	(2.3)
Links (directed)	$L = \sum_i k_i^{\text{in}} = \sum_i k_i^{\text{out}}$	(2.4)
Avg degree (directed)	$\langle k^{\text{in}} \rangle = \langle k^{\text{out}} \rangle = L/N$	(2.5)
Degree distribution	$p_k = N_k/N$	(2.7)
Avg degree via p_k	$\langle k \rangle = \sum_k k p_k$	(2.8)
Degree (undirected, from A)	$k_i = \sum_j A_{ij}$	(2.9)
$k_i^{\text{in}}, k_i^{\text{out}}$ from A	$k_i^{\text{in}} = \sum_j A_{ij}, \quad k_i^{\text{out}} = \sum_j A_{ji}$	(2.10)
Non-zero elements of A	$\sum_{ij} A_{ij} = 2L$	(2.11)
Max links	$L_{\text{max}} = N(N-1)/2$	(2.12)
Weighted adjacency	$A_{ij} = w_{ij}$	(2.13)
Paths of length d	$N_{ij}^{(d)} = [A^d]_{ij}$	(Box 2.4)
Avg path length	$\langle d \rangle = \frac{1}{N(N-1)} \sum_{i \neq j} d_{ij}$	(2.14)
Local clustering coeff.	$C_i = \frac{2L_i}{k_i(k_i-1)}$	(2.15)
Avg clustering coeff.	$\langle C \rangle = \frac{1}{N} \sum_i C_i$	(2.16)
Global clustering coeff.	$C_\Delta = \frac{3 \times \text{triangles}}{\text{triplets}}$	(2.17)